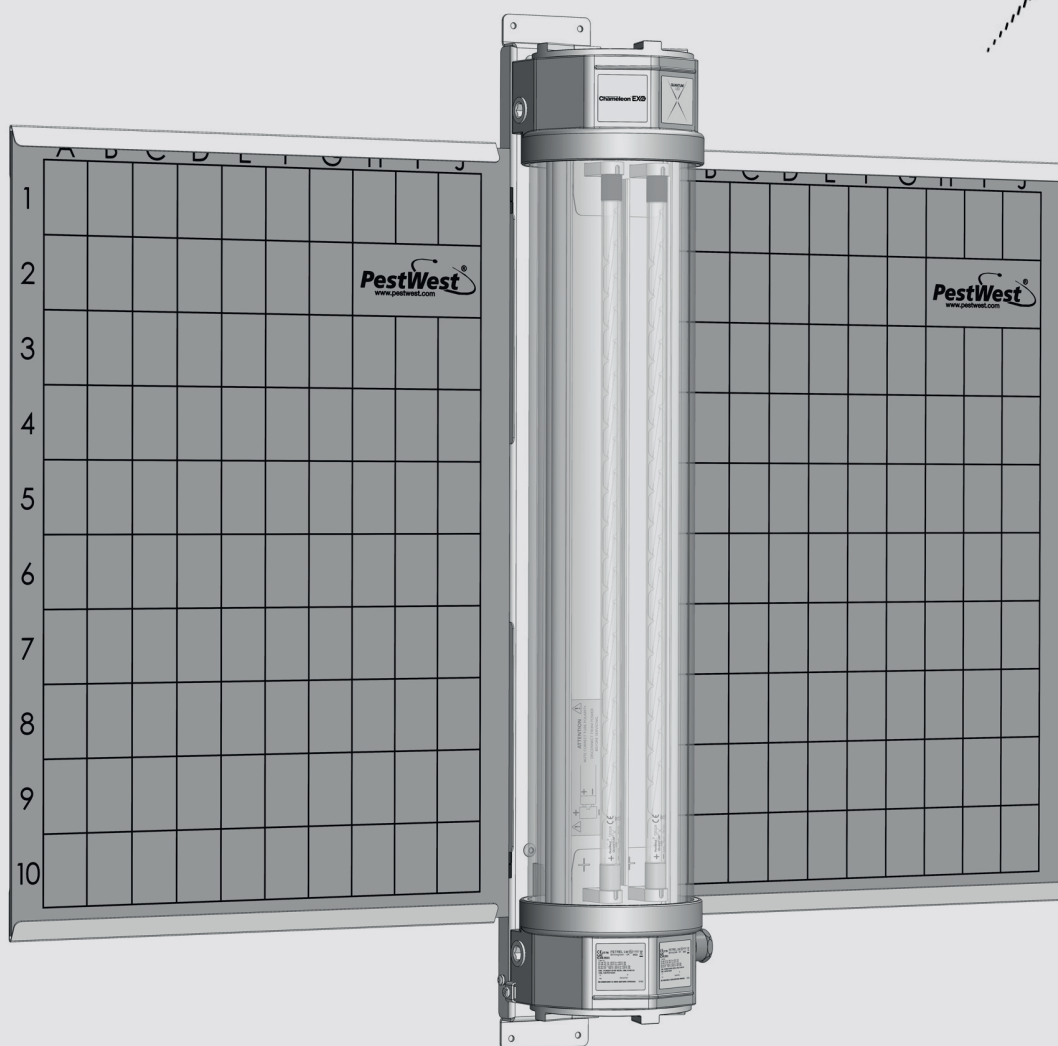




# Chameleon<sup>®</sup> EXG

QUANTUM<sup>®</sup>  
LED



## INSTRUCTIONS FOR USE

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# English

## CHAMELEON EXG X - EXPLOSION-PROOF LED STICKY TRAP

| TECHNICAL SPECIFICATION                        |                                                                                                                  |          |          |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------|----------|
| ATEX Certificate No                            | CML 14ATEX1028X                                                                                                  |          |          |
| ATEX Directive (2014/34/EU)                    | Equipment and protective systems for use in potentially explosive atmospheres                                    |          |          |
| Certification Codes                            | <div> <div>Ex</div> II 2 G - Ex db IIC T6 Gb           <div>Ex</div> II 2 D - Ex tb IIIC T85°C Db         </div> |          |          |
| Ambient Temperature (Ta)                       | -20°C to +55°C                                                                                                   |          |          |
| Ingress Protection Rating                      | IP66                                                                                                             |          |          |
| Operating Zones                                | Gas: Zone 1 & 2 / Dust: Zone 21 & 22                                                                             |          |          |
| Impact Protection                              | IK08: Passed 7 Joules impact test (x3 impacts)                                                                   |          |          |
| UKEX Certificate No                            | CML 21UKEX11018X                                                                                                 |          |          |
| UKEX Statutory requirements (SI 2016 No. 1107) | Equipment and protective systems intended for use in potentially explosive atmospheres regulations 2016          |          |          |
| IECEx Certificate No                           | CML 14.0012X                                                                                                     |          |          |
| European Safety Standards                      | EN IEC 60079-0:2018                                                                                              |          |          |
|                                                | EN 60079-1:2014                                                                                                  |          |          |
|                                                | EN 60079-31:2014                                                                                                 |          |          |
|                                                | EN 60598-1:2021                                                                                                  |          |          |
| Voltage                                        | 220-240V (100-240V Universal input variant available)                                                            |          |          |
| Frequency                                      | 50-60Hz                                                                                                          |          |          |
| Power Input (@ 230V 50Hz)                      | 13.5W                                                                                                            |          |          |
| Dimensions                                     | H: 815mm                                                                                                         | W: 940mm | D: 180mm |
| Weight                                         | 9.7Kg Unboxed                                                                                                    |          |          |
| Mounting                                       | Wall Mounted                                                                                                     |          |          |
| Construction                                   | 304 Grade Stainless Steel / Marine Grade Aluminium / Borosilicate Glass                                          |          |          |
| Coverage                                       | 90-120m <sup>2</sup>                                                                                             |          |          |
| Replacement Tubes                              | 130-000321 - Quantum X LED 370nm 550mm T5 Shatter Resistant                                                      |          |          |
| Replacement Sticky Boards                      | 135-000050 - EXG Sticky Board (6 Pcs)                                                                            |          |          |
| ATEX Cable Gland                               | Hawke 501/421/0 - M20 Thread - 6.5 to 11.9mm Cable Acceptance.                                                   |          |          |

## PRECAUTIONS

All PestWest traps are fully insulated, Class I appliances, nevertheless, sensible precautions should be observed:

- It is essential that the installer checks that the product is suitable for the intended application
- The luminaire must be installed by competent personnel and in accordance with BS EN 60079-14 or equivalent national standards of the country concerned
- No modifications to the luminaire are permitted as this will invalidate certification
- Ensure the unit is properly earthed
- Warning! Do not open when energised
- After de-energising, wait 55 minutes before opening for servicing or maintenance
- Switch off the machine before servicing or cleaning
- The luminaire must not be opened when a potentially explosive atmosphere is present

- Ensure that operators have an adequate and secure working platform, when working on the machine
- Due to the enclosure being made of glass, the equipment should only be installed in areas where there is a low risk of impact
- After 10 years of service life within a potentially hazardous area, relocate the Chameleon EXG X to a non-hazardous area (only in operational condition) Defective tubes must be replaced immediately
- Use only PestWest Quantum X LED 21" tubes (PestWest Part Number 130-000321)
- The Quantum X LED tubes produce ultraviolet light in the UV-A region at 370nm wavelength, which can be harmful to the skin and eyes, avoid prolonged exposure at close proximity
- Do not allow the sticky board to become overfilled with dead insects. Change at regular intervals as required by the catch or at least every 8 weeks
- Peel the release paper from the sticky boards outside of the ATEX zone, as this process may generate static
- The appliance is to be kept out of the reach of children
- For indoor use only

## PRODUCT DESCRIPTION

The Chameleon EXG X has a total anti-corrosive construction including 304 Grade brushed stainless steel, marine grade aluminium alloy and a 7mm thick borosilicate glass enclosure.

The unit is fitted with next generation Quantum X LED filament tubes with a peak wavelength of 370nm coated with FEP shatter-resistant coating as standard.

The new Quantum X LED filament lamp is made with the familiar T5 form factor (glass tube, end caps and pins) used in our conventional 14W T5 fluorescent Quantum tubes.

The main advantages of the Quantum X LED filament lamp technology are the extended pattern of light distribution and the improved energy efficiency. Increased fly attraction is partly due to the LED filaments light distribution, which is around 360° compared with only 120° typical of high power SMD (Surface Mounted Device) LEDs found in older first-generation LED insect light traps. LED filaments are also more energy efficient than both high power SMD UV-A LEDs and also conventional fluorescent tubes.

The Quantum X LED filament lamp consists of 18 individual LED filaments mounted on a frame and hermetically sealed within a glass tube with a state-of-the-art gas cooling system, which is far superior to traditional metal heat sinks.

Each LED filament has 24 individual LEDs which are mounted on a transparent substrate, which enables the Quantum X LED lamp to provide light in all directions. The combination of 360° light distribution and the softer intensity of light produced by far more LEDs (432 per lamp) gives the Quantum X superior fly catching performance.

## RATINGS AND APPROVALS

CE and UKCA marked as well as being certified to the European ATEX Directive 2014/34/EU - Equipment intended for use in potentially explosive atmospheres.

The Chameleon EXG X professional flytrap has been designed for installation in designated hazardous environments and has been tested for compliance to the highest standards applicable within the European community for this grade of product. However, the zoning classification for explosive environments requires proper interpretation and each installation must be judged on its merits. PestWest Electronics are able to provide upon request, copies of independent safety test certificates applicable to this model, to enable a

suitable assessment to be made for a particular installation and no claims other than those therein identified on the certificates are expressed or implied. It is the ultimate responsibility of the installer to ensure that the Chameleon EXG X is suitable for a particular environment. The Chameleon EXG X is suitable for Gas Zones 1 & 2 and Dust Zones 21 & 22. These areas include those where there may be potential for explosive atmospheres in the form of gases and vapours or dust, for example: Alcohol Distillery, Sugar Factories and Flour Mills. The Chameleon EXG X also has an IP66 rating.

UNPACKING YOUR UNIT

Each unit is carefully inspected and packed before leaving our factory. Before discarding the outer carton, examine for obvious evidence of transport damage. Remove the unit carefully and pay particular attention to the removal of all packing materials including any transit packing. Save the packing materials for future use.

MODE OF OPERATION

Electrical flytraps operate on the principal that most flying insects are attracted to light, particularly to the ultraviolet (UV) end of the light spectrum. This light is invisible to humans, however the tubes glow blue which indicates that they are functioning. Since insects are still active and will fly at night (when competition from other light sources is normally lowest), traps should be left operating continuously, i.e: 24 hours a day.

SITING

See 'RATINGS AND APPROVALS' section regarding location within ATEX zones.

The efficiency of the trap can be affected by the intensity of other light sources, for example daylight or other general lighting fixtures. Therefore, mount the trap where competition from other light sources is minimised.

Mount the trap with the intention of attracting flies away from sensitive areas rather than towards them. Observe where the flies tend to congregate and if possible, put your fly trap in or close to that area. Site the trap near where flies are likely to gain access, e.g. doors and windows opening to the outside, especially areas where waste bins are located.

Position the trap where it can be easily reached, ideally around head height for ease of servicing. Do not position the trap directly over working machinery, where access may be difficult or dangerous.

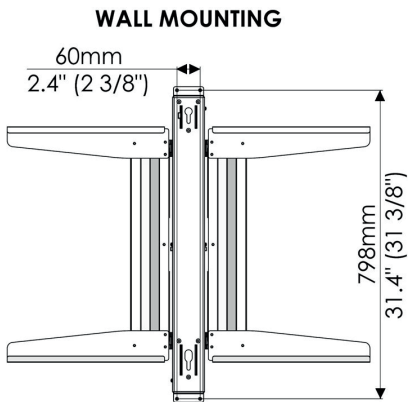
Due to the enclosure being made of glass, the equipment should only be installed in areas where there is a low risk of impact.

INSTALLATION

The Chameleon EXG X must be installed by authorised (and qualified) personnel only. The warranty will be void if the product is modified in any way or used incorrectly. Installation and/or maintenance personnel must check the compatibility (ratings, certifications codes etc.) of this equipment with the environmental and regulatory conditions of the site. Compliance of the rating label with the power supply must be verified. Transportation and storage must be done within the original packing. We recommend storing the Chameleon EXG X in a clean, dry, and aerated environment at a temperature between -10°C and 30°C.

WALL MOUNTING

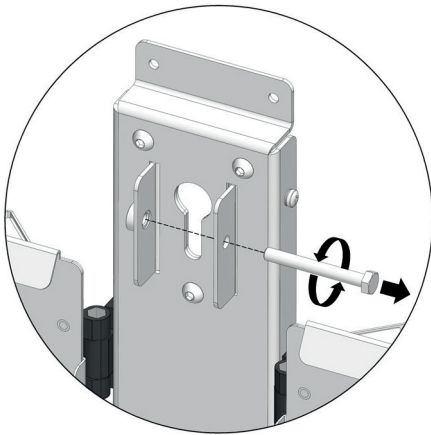
The Chameleon EXG X has 4 holes on the central bracket of the framework to permit wall mounting, using the screws and masonry plugs provided. If the wall is of plasterboard or stud partition etc, suitable blind fixings must be used capable of taking the weight of the unit and to ensure it is securely positioned.



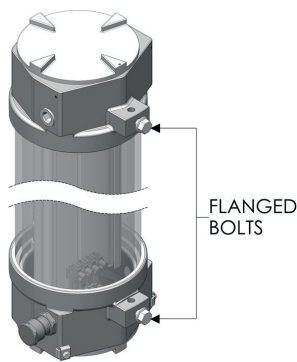
Mark off the fixing points on to the mounting surface using the central bracket as a template. Drill the holes to a depth of 40mm (1 5/8") using a 5.5mm (7/32") drill bit. Place the plastic plugs into the wall, align the central bracket with the plugs and then insert the screws provided and secure the framework in place.

Once the framework has been secured to the mounting surface, the lighting enclosure can then be attached. The enclosure can be mounted with the power cord at either the top or bottom, depending on access restrictions and available power sources.

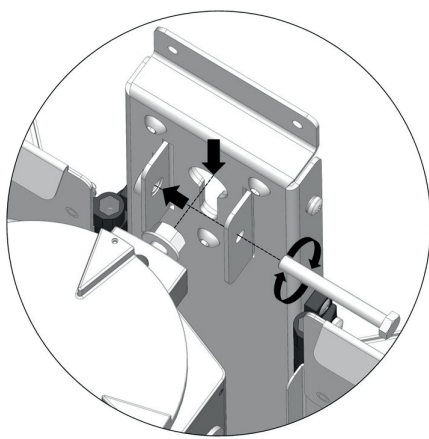
Ensure the retaining bolt has been removed from the top bracket of the framework as shown below.



The Enclosure can then be attached to the framework by slotting the two flanged bolts protruding from the back of the enclosure, into the two keyhole cut-outs in the central bracket of the framework.



The retaining bolt should then be reinserted into the top bracket as shown below, locking the lighting enclosure in position. It is important that the retaining bolt be refitted properly, as it forms part of the units electrical earthing path. Also without it, should the unit be hit or knocked from below, the lighting enclosure could be knocked upwards and out of the fitting.



ELECTRICAL INSTALLATION

**Caution:** Always ensure that the mains supply is safely isolated before carrying out installation or maintenance work.

FITTING THE CABLE GLAND

The enclosure has four M20 threaded cable entry points, three of which come fitted with Ex certified blanking plugs. The enclosure is factory assembled with a plastic blanking plug in one of the entry points which must be removed and replaced with the included M20 Ex certified cable gland. Other alternative configurations can be considered, but care must be taken to make sure the gland(s) and plugs(s) are of a compatible size and rating, and properly fitted.

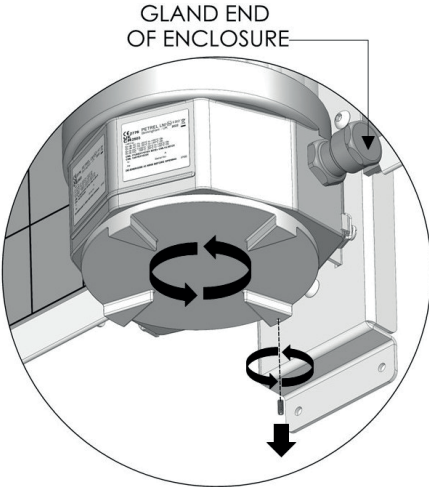
The cable gland(s) must be mounted with lubricated threads using a non-conductive, non-setting grease. Mount the cable gland(s) in such a way as to prevent accidental rotation or loosening.

The Chameleon EXG X comes with one M20 threaded brass cable gland. The manufacturers name and model number can be found in the 'TECHNICAL SPECIFICATIONS' table on Page 2 of this manual. Refer to the cable gland manufacturers User Manual and ATEX certificate for fitting instructions and Specific Conditions of Use. A copy of these can be found on the manufacturers website or can be provided by PestWest upon request.

OPENING THE ENCLOSURE

**Caution:** The luminaire must not be opened when a potentially explosive atmosphere is present. Do not open when energised. After de-energising, wait 55 minutes before opening for servicing or maintenance.

The luminaire is factory assembled to allow removal of the interior gear tray from the Cable Gland end of the enclosure, although access to both ends is available. The orientation of the fitting should therefore be decided prior to installation.



Access to the luminaire is obtained by firstly unscrewing the external locking grub screw using a 1.5mm Allen key. The end cap can now be loosened using a spanner or other suitable tool laid flat across the cover against the 'V' notches and turned anti-clockwise.

Disconnect the floating white plug by squeezing together the side tabs on the plug and pulling the two connectors apart.

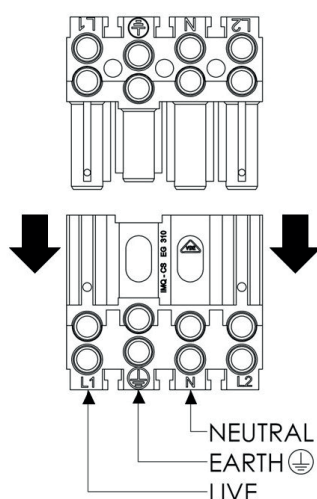
Next pull apart the fixed black mains terminal block from the gear tray.

**⚠ Caution: The black mains terminal block also retains the UV LED lamp holder assembly in place, and if you are entering the enclosure from below, this will now drop down as it is unsupported.**

The lamp holder assembly can now be withdrawn from the enclosure and should be placed on a soft and stable surface with the tubes facing upwards to prevent damage.

### POWER SUPPLY CONNECTIONS

The black mains terminal block will accept up to 4mm<sup>2</sup> wire and should be connected as follows.



Note: L1 is used for switched Live

**⚠ Caution: The power supply cable inside the enclosure should have enough slack in order not to pull on the detachable terminal block.**

### REASSEMBLY OF THE ENCLOSURE

Once the wires have been fitted, the lamp holder assembly can then be reinserted into the enclosure, and the black mains terminal block plugged back into the gear tray. Reconnect the floating white plug and check earth continuity throughout.

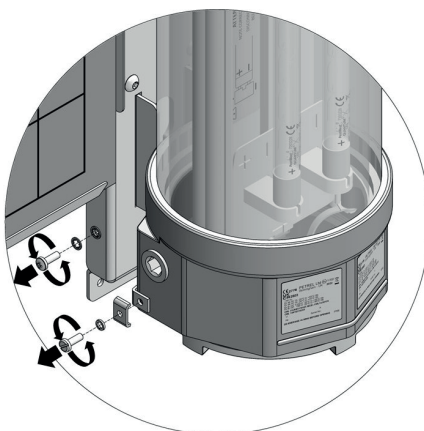
Check that no dirt or dust is in the threads of the end cap or housing. If dirt is present, remove the contaminated grease and any traces of dirt and then reapply new non-conductive, non-setting grease to the threads on both parts.

Inspect the 'O' ring gasket for cracks or damage, and ensure it is correctly seated on refitting.

Replace the end caps by locating the end cap into the threads on the enclosure as to avoid cross threading and potential damage to the threaded flame path. Rotate by hand only in a clockwise motion. The tightening of the end cap can be finished off using a suitable tool as per removal. After replacing the end cap, lightly tighten the locking grub screw to avoid accidental rotation of the end cap.

**⚠ Caution: Excessive tightening of the end cap or grub screw is not required and may damage the threads.**

The Chameleon EXG X lighting enclosure is earth bonded to the Stainless-Steel framework via the flanged bolts protruding from the rear of the enclosure. In certain failure states, such as if the retaining bolt has not been refitted and the enclosure is lifted out of the keyhole slots, this earthing path could be compromised. For added earth bonding protection, the framework and lighting enclosure both have additional earthing points as shown below. To use these, a suitably rated earth cable should be attached between the two parts, being secured and terminated appropriately.



The framework has two earthing points, one on the lower left side, and another on the upper right side. This means whichever orientation the enclosure is used (Cable Gland end facing up or down), there will be a suitably positioned earthing point available.

### CHANGING THE STICKY BOARDS

**⚠ Caution: Peel the release paper from the sticky boards outside of the ATEX zone, as this process may generate static.**

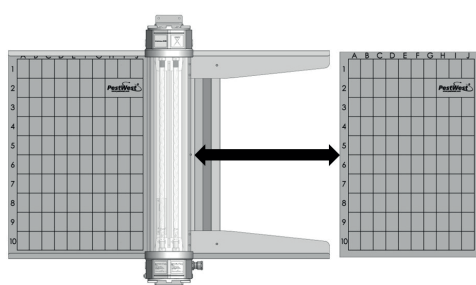
Always install PestWest sticky boards to ensure peak performance. This sticky board has been developed in conjunction with our Quantum X LED technology to provide optimum fly attraction and retention. Using any other sticky board will have a negative impact on the performance of the system.

Sticky boards should be changed when full, or at least every six to eight weeks. Simply remove the used sticky boards from the machine by withdrawing them sideways out of the unit.

Should it be necessary to retain the boards for reference purposes, they may be dated and covered in clingfilm to facilitate storage. This should occur outside of the ATEX zone due to risk of static.

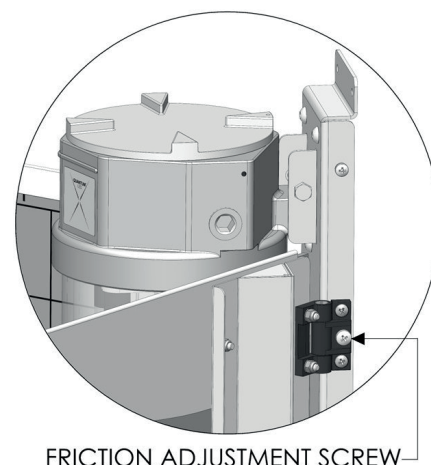
Peel the release paper from the new sticky boards whilst outside of the ATEX zone, as again this process may generate static. The release paper should be removed in a single swift motion, otherwise they will tend to remain firmly attached. Should this occur, re-apply the peeled back section and repeat.

Insert the new glue boards between the runners at the top and bottom either side of the unit as shown below, ensuring that the glue surface is facing outwards and that the boards are pushed fully home. The change is now complete.



### STICKY BOARD ANGLE ADJUSTMENT

The Chameleon EXG X is fitted with moveable sticky board holders which can be positioned closer to the UV light source to improve the catch performance of the machine. This feature is also useful when installing the unit where wall space is reduced or there are obstructions. The sticky board holders are mounted on adjustable friction hinges and in environments where there are drafts present, it may be necessary to stiffen this movement. This can easily be achieved by tightening the central friction adjustment screws on the black hinges, turning in a clockwise motion using a pozi-drive screwdriver as shown below.

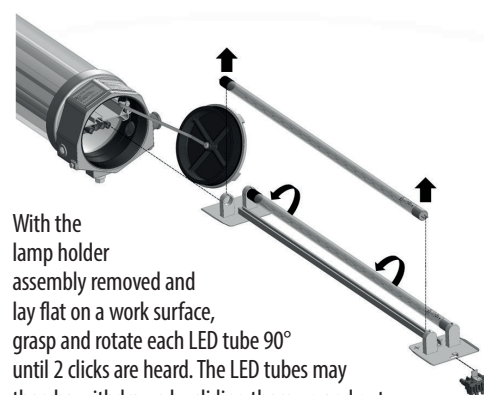


### CHANGING THE UV LED TUBES

**⚠ Caution: Always ensure that the mains supply is safely isolated before carrying out installation or maintenance work.**

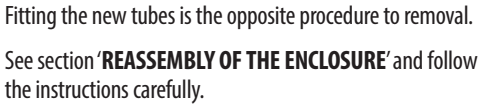
See section 'OPENING THE ENCLOSURE' and follow the instructions carefully.

The unit will only operate with PestWest Quantum X LED 370nm 550mm T5 tubes.



With the lamp holder assembly removed and lay flat on a work surface, grasp and rotate each LED tube 90° until 2 clicks are heard. The LED tubes may then be withdrawn by sliding them up and out of the lamp holders.

**⚠ Caution – Quantum X LED tubes are powered by Direct Current (D.C.) at one end only, which is easily recognised by the silver end cap and the printed + & - symbols etched onto the glass. The D.C. input pins require fitting in the correct polarity, positive (+) and negative (-). Please note that the correct polarity is marked with a + & - symbol on the Quantum X LED glass tube and the corresponding pins must be fitted to the corresponding + & - symbols on the metalwork, located on the powered end, next to the lamp holders as shown in the following diagram.**



**⚠ Caution:** Always ensure that the mains supply is safely isolated before cleaning and maintenance.

Threaded flame paths should be checked periodically to ensure that they are in good condition and free from dirt, dust and corrosion.

Ensure that the flame paths are treated with non-conductive, non-setting grease before re-sealing.

## GUARANTEE

\* Manufacturer reserves the right to vary model specifications without notice.

[illegible]



Specialist Range



**PestWest Electronics Ltd operates a quality management system certified to ISO 9001**

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Confidence in  
**Fly Control**